IG02

Climate of India

About 1-2 questions a paper. Documented repeats: the retreating monsoon on the Tamil Nadu coast (RAS 2023 and 2024), rainfall variability highest in west Rajasthan and Ladakh (RAS 2024), and the Chennai-versus-Mumbai rainfall contrast (RAS 2024).

One chapter, one idea: the monsoon. Understand why the winds reverse and the rest of the chapter falls into place - onset dates, rain-shadows, variability, even the Tamil Nadu anomaly. RPSC does not ask you to explain the monsoon. It asks you for the fact that follows from the explanation, so learn the mechanism once and then drill the facts.

Climate of India

Mechanism
- Differential heating
- ITCZ shifts north
- Coriolis deflection
- Tibetan plateau heating

Jet streams
- Subtropical westerly
- Tropical easterly
- Somali (Findlater) jet

Four seasons (IMD)
- Winter Jan-Feb
- Pre-monsoon Mar-May
- SW monsoon Jun-Sep
- Post-monsoon Oct-Dec

Winter rain
- Western disturbances
- Mediterranean origin
- Mahawat
- Rabi wheat

Onset and withdrawal
- Kerala 1 June
- Whole country 8 July
- West Rajasthan 17 September
- Complete by 15 October

Retreating monsoon
- October heat
- Coromandel rain Oct-Dec
- Bay cyclones

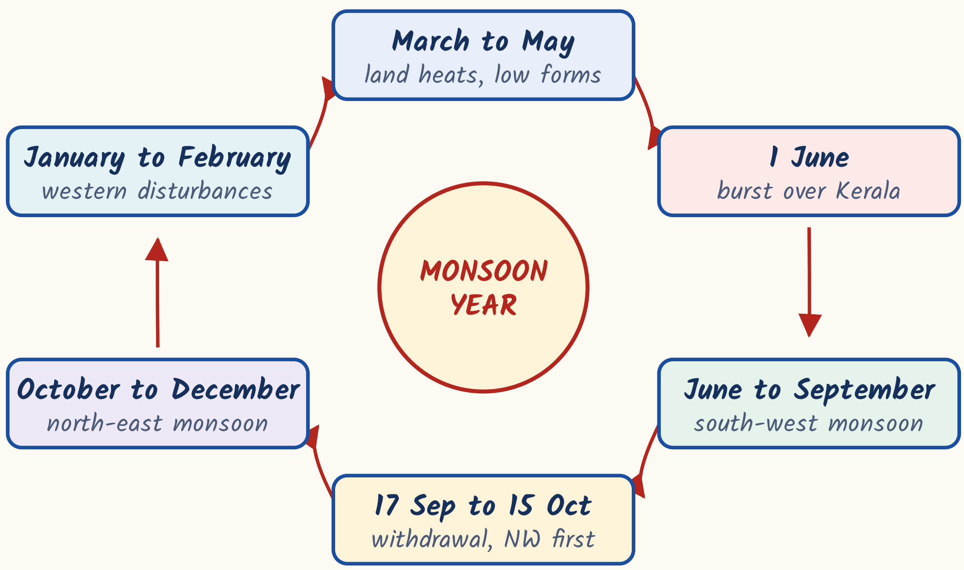
Swing factors
- El Nino
- La Nina
- IOD
- SOI Tahiti-Darwin
- MONEX

Koppen types
- Amw west coast
- As Coromandel
- Aw peninsula
- BWhw west Rajasthan
- Cwg Ganga plain

What a monsoon is, and the machinery that drives it

A monsoon is a seasonal reversal of wind direction. In summer the land heats faster than the sea, a low pressure area forms over north-west India, and moist air is drawn in from the ocean. In winter the land cools faster, pressure rises over the interior, and dry air blows out towards the sea. That is the old, simple, thermal explanation, and it is still the first thing to say.

The monsoon year – one full reversal



But heating alone does not explain why the rain arrives in a sudden burst in early June. The modern explanation adds three things in the upper air. One, the ITCZ – the equatorial low pressure trough – shifts north in summer and sits over the Ganga plain as the monsoon trough. Two, the south-east trade winds of the southern hemisphere cross the equator, get deflected to the right by the Coriolis force, and arrive over India as south-west winds. Three, the jet streams change position. Until the subtropical westerly jet moves out of the way, the monsoon cannot break.

The three jet streams and the trough - learn one line each

Wind system	What it does
Subtropical westerly jet	Lies south of the Himalaya in winter; shifts north of the Tibetan plateau in early June - a precondition for the burst
Tropical easterly jet	Develops over peninsular India in summer, above the intensely heated Tibetan plateau
Somali (Findlater) jet	Low-level cross-equatorial jet off the East African coast; strengthens the Arabian Sea branch
Monsoon trough	The northward-shifted ITCZ lying over the Ganga plain in July

Why the monsoon bursts in early June

Land heats, low pressure deepens over north-west India → Subtropical westerly jet withdraws north of the Tibetan plateau →

Tropical easterly jet establishes over the peninsula → Somali jet strengthens the Arabian Sea branch →

South-east trades cross the equator and turn south-west → Burst of the monsoon over Kerala, about 1 June

YAAD RAKHO

Two jets, two directions, two seasons. Westerly jet = winter feature, brings western disturbances. Easterly jet = summer feature, sits over the peninsula. If a question puts western disturbances with the easterly jet, it is wrong.

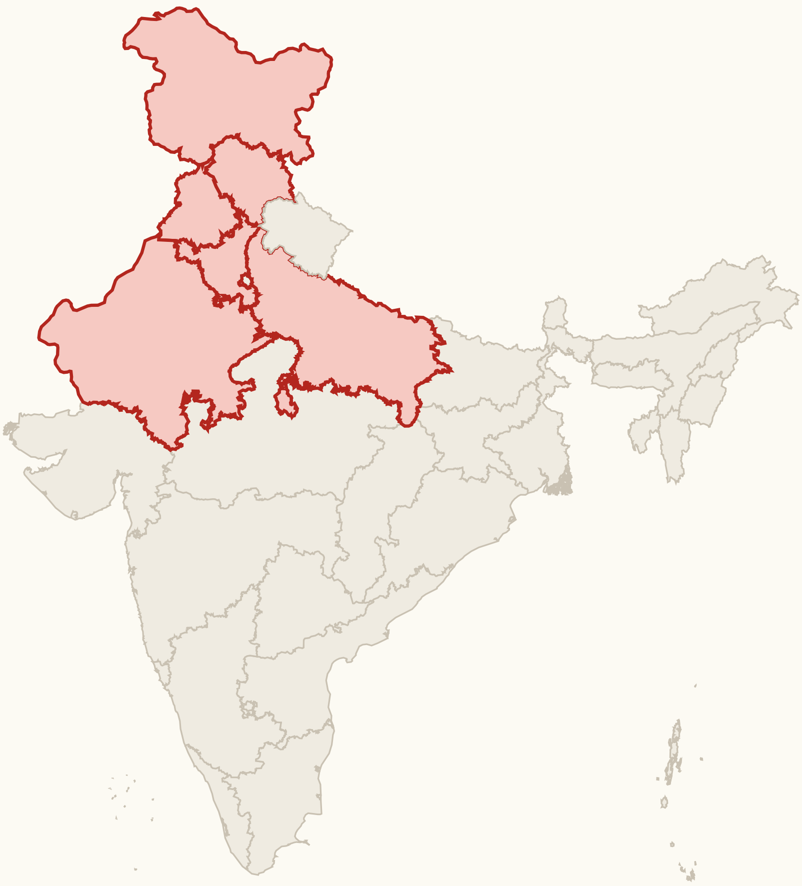
CURRENT LINK

Current link (as of September 2026): the Bharat Forecast System was launched on 26 May 2025 by the Ministry of Earth Sciences. Developed at the Indian Institute of Tropical Meteorology, Pune, it runs on a 6 km grid - the finest resolution of any operational weather model in the world at launch, against the 12 km grid it replaced - and is meant to take forecasting down to panchayat level, which matters most for monsoon rainfall and cyclone warnings. Carry three facts: 2025, 6 km, IITM Pune.

Winter, and the western disturbances that feed the rabi crop

Winter in India should be bone dry, because the winds blow from land to sea. Yet Punjab, Haryana, western UP and northern Rajasthan get a little rain in December to February, and that small rain decides the wheat crop. The reason is the western disturbance - an extra-tropical cyclone that forms over the Mediterranean region and is dragged across Iran, Afghanistan and Pakistan into India by the subtropical westerly jet stream.

The winter rain belt of the western disturbances



Mahawat here decides the rabi wheat; snow in the western Himalaya

- Western disturbances are extra-tropical cyclones of Mediterranean origin.
- They are steered into India by the subtropical westerly jet, not by any monsoon wind.
- The winter rain they bring is locally called mahawat.
- Mahawat is small in amount but decisive for the rabi wheat crop.
- They also cause snowfall in the western Himalaya and occasional hailstorms in Rajasthan.
- Their track weakens as they move east, so eastern India gets almost nothing from them.

IMD's four seasons

Season	Months
Winter	January and February

Season	Months
Pre-monsoon / hot weather	March to May
South-west monsoon	June to September
Post-monsoon / retreating monsoon	October to December

TRAP
Western disturbances originate over the Mediterranean, not the Arabian Sea and not the Bay of Bengal. And they are carried by the subtropical westerly jet, not the tropical easterly jet. RPSC has used both of these as the wrong option in a 'NOT correct' question.

- Say the chain as one phrase - Mediterranean, westerly jet, mahawat, wheat.
- Four words, one chain, and three separate RAS questions answered.

Hot weather season and the local winds

March to May the interior bakes. Temperature rises, pressure falls, and before the monsoon arrives the country gets a scatter of local winds and thunderstorms with regional names. These names are pure matching material - learn wind to region, nothing else.

Local winds and pre-monsoon showers

Name	Region
Loo	Northern plains, May-June - hot dry westerly
Kal Baisakhi	West Bengal - violent pre-monsoon nor'wester
Bardoli Chheerha	Assam - the nor'wester of the Brahmaputra valley
Mango showers	Kerala and coastal Karnataka
Blossom showers	Coffee areas of Karnataka

TRAP

Bardoli Chheerha belongs to Assam, not Punjab. That exact wrong pair has been used in a 'NOT correctly matched' question. Kal Baisakhi is Bengal, Bardoli Chheerha is Assam - two nor'westers, two different states.

- Mango showers help the mango crop ripen in Kerala and coastal Karnataka.
- Blossom showers help coffee flowers to set in Karnataka.
- The Loo is a heat wind, not a rain-bearing wind.

Onset, advance and withdrawal - the calendar

The monsoon does not arrive everywhere at once. It bursts on the Kerala coast, splits into two branches around the peninsula, and sweeps north-west over about five weeks. It then retreats in the reverse direction - out of the north-west first, out of the far south last. Get the direction of advance and retreat right and the dates follow.

Onset and advance of the south-west monsoon

Kerala coast - normal onset 1 June → Splits into the Arabian Sea branch and the Bay of Bengal branch → Arabian Sea branch strikes the Western Ghats; Bay branch is turned west by the Himalaya → The two branches converge over the north-western plains → Whole country covered by about 8 July

Withdrawal – the reverse order

Begins from west Rajasthan – IMD revised normal date 17 September → Retreats from the north-western plains through late September → Clears the Ganga plain and central India in October → Withdrawal from the whole country complete by about 15 October → Winds now blow off the land towards the Bay – the north-east monsoon

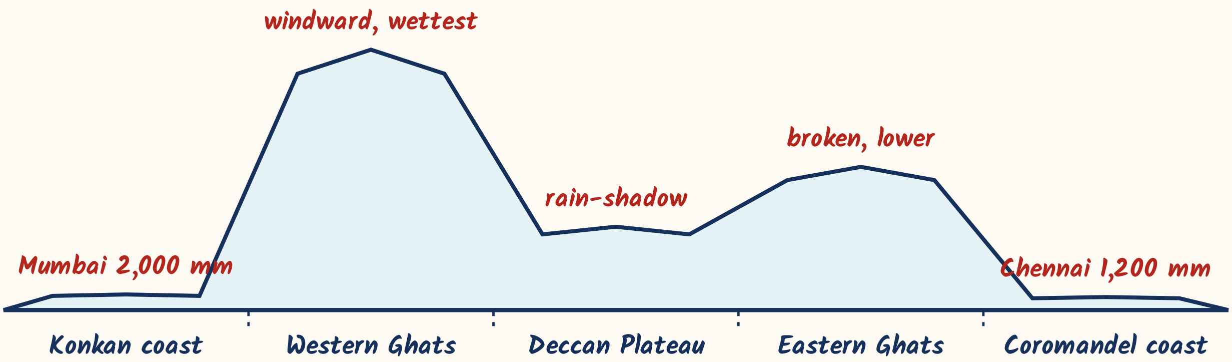
- IMD revised the normal start of withdrawal from west Rajasthan from 1 September to 17 September.
- The Arabian Sea branch gives heavy rain on the windward western slopes of the Western Ghats and leaves a rain-shadow on the Deccan plateau.
- Mawsynram in the Khasi hills of Meghalaya has the world's highest average annual rainfall; funnel-shaped valleys trap the Bay branch.
- A 'break' in the monsoon is a dry spell within the rainy season, caused by a shift in the position of the monsoon trough.
- West Rajasthan (Jaisalmer, Barmer) and Ladakh receive less than 20 cm of rain; Leh is a cold desert.

- Advance is south-east to north-west; withdrawal is north-west to south-east.
- The monsoon leaves first from where it arrived last – that one line answers the direction question in either form.

The retreating monsoon and the Tamil Nadu coast – the biggest repeat here

This is the part of the chapter RPSC comes back to. Tamil Nadu is the odd one out in Indian climate. It stays dry when the rest of India is drenched, and it gets its rain when the rest of India is drying out. Two reasons, and you must be able to give both. First, the Coromandel coast lies in the rain-shadow of the Western Ghats, so the Arabian Sea branch has already lost its moisture by the time it crosses. Second, the coast runs parallel to the Bay of Bengal branch instead of standing across it, so that branch does not strike it either.

West to east across the peninsula – why Mumbai is wet and Chennai is not



Then in October the winds reverse. The retreating north-east monsoon blows from land to sea, but on its way to Tamil Nadu it crosses the Bay of Bengal, picks up moisture, and delivers it on the Coromandel coast in October to December. That is why Chennai's rainiest months are October and November.

ASKED IN RAS

Asked in RAS Pre 2023 and again in 2024 – the Tamil Nadu coast receives the bulk of its annual rainfall in October to December from which system? Answer: the retreating or north-east monsoon.

ASKED IN RAS

Asked in RAS Pre 2023, also framed as assertion-reason - why is the Coromandel coast dry during the south-west monsoon? Answer: it lies in the rain-shadow of the Western Ghats and runs parallel to the Bay of Bengal branch.

Mumbai vs Chennai - the contrast RPSC asked in 2024

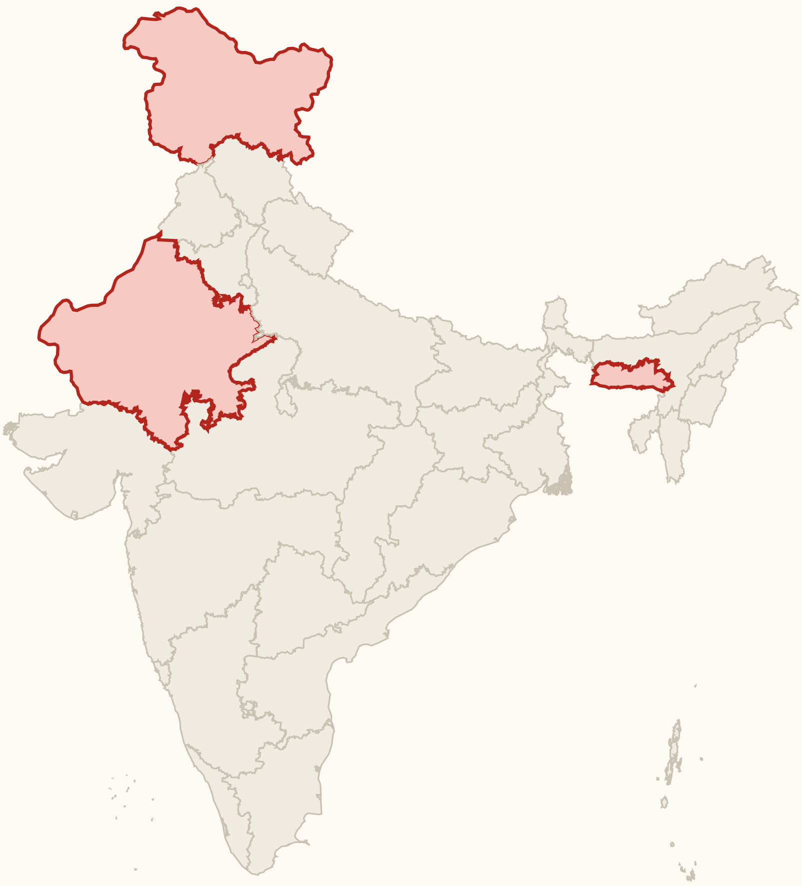
Mumbai (west coast)	Chennai (Coromandel coast)
Over 2,000 mm a year	About 1,200 to 1,400 mm a year
Rain concentrated June to September	Maximum rain in October and November
Fed by the Arabian Sea branch of the SW monsoon	Fed mainly by the retreating north-east monsoon
Windward side of the Western Ghats	Rain-shadow side of the Western Ghats
Low rainfall variability, below 25 per cent	Higher variability, dependent on a short season

- The retreating monsoon season brings the oppressive 'October heat' - high temperature with high humidity and clear skies.
- Tropical cyclones are frequent over the Bay of Bengal in this season.
- The withdrawal of the monsoon begins from north-western India, not from the south.

Rainfall distribution, variability and Koppen's regions

One rule explains the whole variability map: variability is inversely related to amount. Where rainfall is heavy it is also dependable; where rainfall is scanty it is also erratic. That is why west Rajasthan, which gets least, also suffers the wildest swings - and why droughts are a Rajasthan problem and not a Kerala problem.

The two extremes of the same rainfall map



Mawsynram wettest on earth; west Rajasthan and Ladakh under 20 cm

- Variability above 50 per cent - west Rajasthan, Ladakh, the interior Deccan.
- Variability below 25 per cent - the west coast, the north-east, the Himalaya.
- Rainfall variability is inversely related to rainfall amount.
- Mawsynram highest, west Rajasthan and Ladakh lowest - the two extremes of the same map.

ASKED IN RAS

Asked in RAS Pre 2024 - the coefficient of variability of annual rainfall in India is highest in which region? Answer: western Rajasthan and Ladakh. Note the Rajasthan angle; RPSC will nearly always pick the option that carries the state's name when the fact allows it.

Koppen's climatic types in India

Code	Type and region
Amw	Monsoon with short dry season - the west (Malabar) coast
As	Monsoon with dry summer - the Coromandel coast
Aw	Tropical savanna - most of the peninsula
BShw	Semi-arid steppe
BWhw	Hot desert - west Rajasthan
Cwg	Monsoon with dry winter - the Ganga plain
Dfc	Cold humid winter with short summer - Arunachal Pradesh
E	Polar - the high Himalaya

TRAP
Arunachal Pradesh is Dfc, not Aw. Aw is the tropical savanna of the peninsula. Also keep As (Coromandel, dry summer) apart from Amw (west coast, short dry season) - the two coasts, two codes, and RPSC has matched them both.

What swings the monsoon, and the hazards it leaves behind

The monsoon's strength changes from year to year, and the drivers sit in the oceans. El Nino is an abnormal warming of the eastern Pacific, and it usually goes with a weak Indian monsoon. La Nina is the opposite phase and usually goes with a strong one. The Indian Ocean Dipole works closer to home: when the western Indian Ocean is warmer than the east, the dipole is positive and the monsoon does well - strongly enough at times to cancel out an El Nino.

- El Nino - warm eastern Pacific - weak or deficient Indian monsoon.
- La Nina - above-normal monsoon rainfall over India.
- ENSO is tracked by the Southern Oscillation Index - the pressure difference between Tahiti and Darwin.
- Positive Indian Ocean Dipole - warmer western Indian Ocean - favours a good monsoon and can offset an El Nino.
- MONEX, the Monsoon Experiment, ran under the Global Weather Experiment.
- Winter MONEX December 1978; Summer MONEX May to July 1979, over the Arabian Sea and Bay of Bengal.

Cyclones, droughts, floods - the hazard facts

Item	Fact
Bay vs Arabian Sea cyclones	Bay of Bengal cyclones are about four times as frequent
Most cyclone-prone coast	The Odisha-Andhra Pradesh coast
Cyclone peaks	Just before and just after the south-west monsoon
Naming authority	IMD as the RSMC New Delhi, for thirteen WMO/ESCAP member countries
Moderate drought	Rainfall deficiency of 26 to 50 per cent of normal
Severe drought	Rainfall deficiency of more than 50 per cent of normal
Most flood-prone belts	Brahmaputra valley, the Kosi belt of north Bihar, the Damodar basin

TRAP

A deficiency of 26 to 50 per cent is a moderate drought - not 10 to 25 per cent. RPSC has planted the 10-25 figure as the wrong statement in a statement-based question. Learn the two thresholds as 26 and 50.

CURRENT LINK

Current link (as of September 2026): IMD, founded in 1875 and headquartered at New Delhi, completed 150 years on 15 January 2025, and Mission Mausam was launched around that anniversary to upgrade India's weather observation and modelling. Expect the founding year 1875, the 150-year milestone and Mission Mausam to appear as a current-affairs one-liner.

REVISION RECAP

1. Monsoon = seasonal reversal of winds; thermal low over north-west India plus the northward shift of the ITCZ.
2. South-east trades cross the equator and are deflected right by the Coriolis force to become south-west monsoon winds.
3. Subtropical westerly jet must withdraw north of the Tibetan plateau before the monsoon can burst.
4. Tropical easterly jet is a summer feature over the peninsula; the Somali (Findlater) jet strengthens the Arabian Sea branch.
5. Western disturbances - extra-tropical cyclones of Mediterranean origin, carried by the westerly jet; their rain is mahawat and it saves the rabi wheat.
6. IMD seasons: winter Jan-Feb, pre-monsoon Mar-May, south-west monsoon Jun-Sep, post-monsoon Oct-Dec.
7. Onset over Kerala 1 June; whole country covered by about 8 July.
8. Withdrawal begins from west Rajasthan on the revised normal date of 17 September; complete by about 15 October.
9. Arabian Sea branch soaks the windward Western Ghats and leaves the Deccan in rain-shadow; Mawsynram records the world's highest average rainfall.
10. Tamil Nadu is dry in the south-west monsoon - rain-shadow plus a coast parallel to the Bay branch.
11. The retreating north-east monsoon waters the Coromandel coast in October-December; October heat marks the season.
12. Mumbai over 2,000 mm in June-September; Chennai 1,200-1,400 mm peaking in October-November.
13. Variability is inversely related to amount - above 50 per cent in west Rajasthan and Ladakh, below 25 per cent on the west coast.
14. El Nino weakens the monsoon, La Nina strengthens it, a positive IOD helps and can offset El Nino; SOI = Tahiti minus Darwin.

15. MONEX 1978-79 studied the monsoon under the Global Weather Experiment.
16. Koppen: Amw west coast, As Coromandel, Aw peninsula, BWhw west Rajasthan, Cwg Ganga plain, Dfc Arunachal, E high Himalaya.
17. Bay cyclones about four times as frequent as Arabian Sea ones; Odisha-Andhra coast most vulnerable; IMD names them as RSMC New Delhi.
18. Moderate drought = 26-50 per cent deficiency; severe = above 50 per cent; IMD founded 1875.